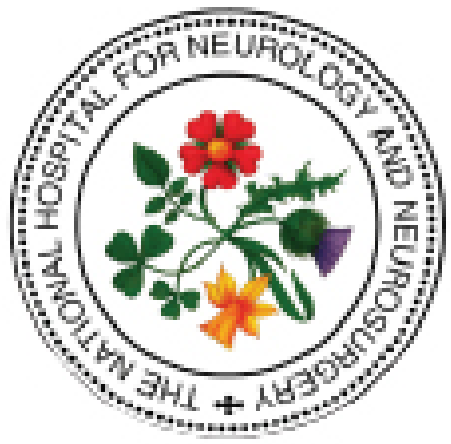




FLUID BALANCE and FLUID CHALLENGE for PATIENTS with ACUTE BRAIN INJURY

FLUID BALANCE

HOW MUCH FLUID?

- **EUVOLAEMIA** is target for most patients with acute brain injury
- Subarachnoid haemorrhage (SAH) patients traditionally have 3L/24h but consider risks in the elderly and in patients with cardiac history
- Agree daily fluid balance target for patient

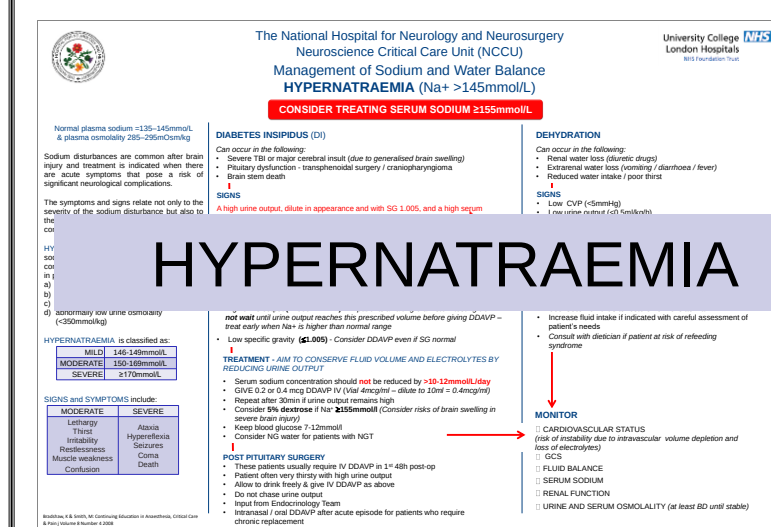
WHICH FLUID?

IMPORTANCE OF FLUID TYPE AND TONICITY

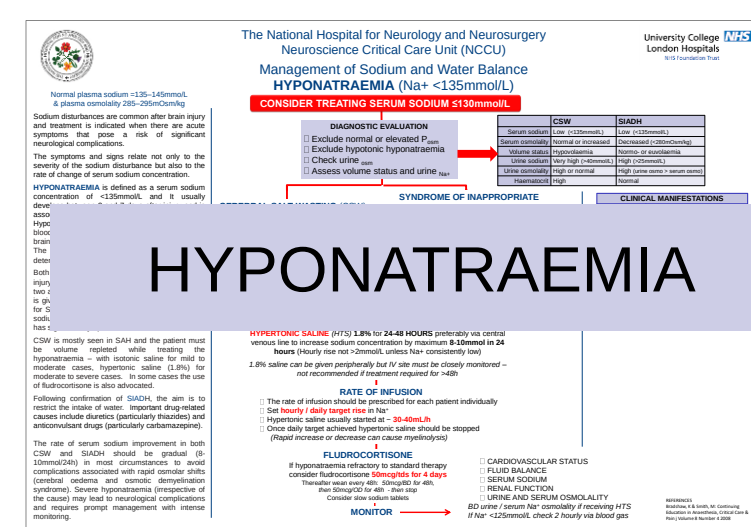
- Isotonic fluids
- 1st choice: **Sodium Chloride 0.9%**
- Avoid hypo-osmolar solutions (*colloids used only in resuscitation scenario*)
- Avoid glucose containing solutions
- Monitor electrolytes

URINE OUTPUT

- Normal range: 0.5-2ml/kg/h
- If output high consider whether result of excessive IV fluids or Hyperosmolar therapy
- Consider disorders of sodium and water balance (*see protocols*)



HYPERNATRAEMIA



HYPONATRAEMIA

FLUID CHALLENGE

REDUCED PLASMA VOLUME SUSPECTED?

Hypotension, tachycardia, oliguria - **Apply FloTrac**

- ✓ Systolic BP <100mmHg
- ✓ Heart rate >90bpm
- ✓ Capillary refill >2s
- ✓ Peripheries cold to touch
- ✓ Respiratory rate >20bpm
- ✓ NEWS ≥5
- ✓ 45° passive leg raising - suggests fluid responsiveness



FloTrac Transducer NOT MRI compatible - please change to regular arterial line transducer in preparation for MRI

Check fluid balance - Last 24h + cumulative

Consider **250mL** fluid challenge IV crystalloid over **10 minutes**
MEASURE SV

In patients with cardiac history or elderly patients consider **100ml bolus** and repeat as tolerated

10% increase in SV or CLINICAL IMPROVEMENT?

YES

NO

STOP
REVIEW
FLUID
BALANCE
& MAKE PLAN

REPEAT TO
MAXIMUM OF
3 CHALLENGES
- review responsiveness after each challenge

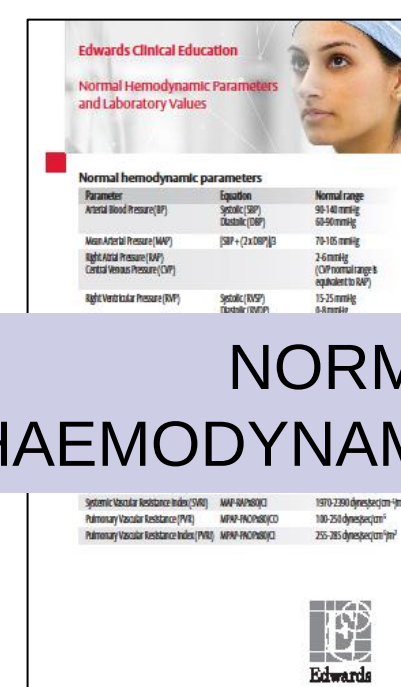
10% increase in SV or CLINICAL IMPROVEMENT?

YES

NO

STOP
REVIEW
FLUID
BALANCE
& MAKE PLAN

CONSIDER
OTHER
CIRCULATORY
SUPPORT



NORMAL HAEMODYNAMIC VALUES

FLUID RESUSCITATION

- Unless evidence of intravascular volume overload give 500mL IV fluid over 15 min
- Reassess patient to determine need for more fluid - *Document if IV fluid contraindicated*
- Fluid responsiveness – check resolution of adverse signs *hypotension, tachycardia, altered consciousness, oliguria, hyperlactatemia*
- Crystalloids (Hartmann's solution - but not 5% glucose) appropriate choice of fluid - *any colloid used should be limited to initial resuscitation phase*
- Blood products may be required to maintain normal coagulation or haemoglobin level ≥70 g/L

